# Electric Vehicle Market Performance & Insights

Academic Project Report by Manshi Kourav (Enrollment No. 0103AD241116, LNCT Bhopal) | SkillWallet | Data Analytics with Tableau

## 1. Cover Page

| | |
|---|---|
| **Project Title** | Electric Vehicle Market Performance & Insights |
| **Document Type** | Academic Project Report |
| **Student Name** | Manshi Kourav |
| **Enrollment No.** | 0103AD241116 |
| **College** | LNCT Bhopal |
| **Project Domain** | Data Analytics |
| **Module** | Data Analytics with Tableau |
| **GitHub Repository** | https://github.com/manshikourav04/EV-Market-Analysis |
| **Live Application** | https://manshikourav04-ev-market-analysis-app-twyxno.streamlit.app |
| **Main Application File** | app.py (branch: main) |

> **Note on report integrity.** This report separates *confirmed implementation* from *proposed methodology* and *pending work*. Where evidence is not yet available, a clearly labelled placeholder such as [INSERT ACTUAL DATASET SOURCE] is used instead of assumed information.

## 2. Certificate

This is to certify that **Manshi Kourav** has carried out the project work titled **"Electric Vehicle Market Performance & Insights"** in the domain of Data Analytics, as part of the module **Data Analytics with Tableau**, under the SkillWallet programme.

The work presented in this report is the candidate's own and has been prepared in accordance with academic requirements.

| | |
|---|---|
| **Institute / Organisation** | LNCT Bhopal |
| **Mentor / Guide Name** | [INSERT MENTOR NAME] |
| **Date** | [INSERT DATE] |
| **Mentor Signature** | _____ |
| **Student Signature** | _____ |

## 3. Declaration

I, **Manshi Kourav**, hereby declare that the project report titled **"Electric Vehicle Market Performance & Insights"** is a record of work carried out by me for the module Data Analytics with Tableau. This report has not been submitted, in whole or in part, to any other institution or programme for any degree, diploma, or certification.

I further declare that:

- All sources of information, tools, and documentation used in this project have been acknowledged.
- No data, results, screenshots, or findings have been fabricated. Items that could not be verified at the time of writing are marked as pending.
- I have followed the principles of academic integrity throughout the preparation of this project and report.

**Enrollment No.:** 0103AD241116
**College:** LNCT Bhopal

**Student Name:** Manshi Kourav
**Signature:** _____
**Date:** [INSERT DATE]

## 4. Acknowledgement

I express my sincere gratitude to SkillWallet for providing the learning platform and the project opportunity that made this work possible. I thank my mentor, [INSERT MENTOR NAME], for guidance, feedback, and encouragement during the project.

I am also grateful to the open-source communities behind Python, Pandas, NumPy, and Streamlit, and to the teams who maintain the official documentation of these tools, as well as that of Tableau and GitHub. Their freely available resources were essential for learning and for building the application. Finally, I thank my family and friends for their continued support.

## 5. Abstract

Electric vehicles (EVs) are an increasingly important part of the global automobile industry, driven by concerns about emissions, fuel costs, and advances in battery and charging technology. Understanding how the EV market performs requires structured analysis of large volumes of data covering vehicles, their characteristics, and market-related attributes.

This project, **Electric Vehicle Market Performance & Insights**, applies data analytics to explore EV market data and present it through a web-based application. A Python application built with Streamlit (main file `app.py`) has been developed, hosted in a public GitHub repository, and deployed online. The application is titled *Electric Vehicle Market Analysis* and *EV Market Trends and Insights*. On loading, it displays the message "Datasets loaded successfully!" together with a dataset overview of **200,000 training records**, **100,000 testing records**, and **37 columns**.

The module for this project, Data Analytics with Tableau, also anticipates Tableau-based visualisation. At the time of writing, no evidence of Tableau dashboards or stories, or of their integration with the Streamlit application, has been supplied; these are therefore documented as a proposed workflow pending verification. The report describes the confirmed implementation, the recommended methodology for data preparation, exploratory analysis and visualisation, testing approaches, limitations, and future enhancements. No analytical findings are claimed beyond what has been verified.

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*Tip: in the editor, heading styles can be used to generate or refresh a navigable outline.*

# 7. Introduction

## 7.1 Background of Electric Vehicles

An electric vehicle uses one or more electric motors for propulsion, powered by energy stored in a rechargeable battery rather than by burning fuel in an internal combustion engine. Categories commonly discussed in the industry include battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs). Compared with conventional vehicles, EVs produce no tailpipe emissions and have fewer moving parts in the drivetrain, which has made them a focus of environmental and transport policy.

## 7.2 Growth of the EV Industry

Over the past decade, the EV industry has attracted significant attention from manufacturers, governments, investors, and consumers. Factors frequently cited as drivers of this growth include improvements in battery technology, expanding charging infrastructure, government incentives and emission regulations, and rising consumer

interest in sustainable transport. Because this report does not rely on externally cited statistics, no specific market-size or growth figures are quoted here; readers should consult official industry sources for such data.

### 7.3 Importance of Data Analytics in the Automobile Industry

The automobile industry generates data at every stage, from design and manufacturing to sales, ownership, and usage. Data analytics helps stakeholders to:

- Identify patterns and trends in vehicle characteristics and market behaviour.
- Compare the performance of different segments, makes, or models.
- Support planning decisions such as product positioning, infrastructure planning, and policy evaluation.
- Communicate findings to non-technical audiences through visual dashboards.

Tools such as Python, Pandas, Streamlit, and Tableau allow analysts to move from raw data to interactive, shareable insight.

### 7.4 Purpose of the Project

The purpose of this project is to apply data analytics techniques to an EV market dataset and present the data in an accessible web application. The project also serves as a learning exercise in the complete analytics lifecycle: data handling, analysis, visualisation, application development, deployment, and documentation.

## 8. Problem Statement

### 8.1 Need to Analyse EV Market Data

As EV adoption grows, the volume and variety of data describing vehicles and markets also grow. Raw datasets of this size, such as one containing hundreds of thousands of records and dozens of columns, cannot be understood through manual inspection. Systematic analysis is needed to organise the data, understand its structure, and extract information that supports decisions.

### 8.2 Challenges in Understanding Market Trends and Performance

- **Scale:** large datasets are difficult to inspect without programmatic tools.
- **Dimensionality:** many columns make it hard to identify which attributes matter.
- **Data quality:** real-world data commonly contains missing, duplicate, or inconsistent values that must be checked before analysis.

- **Communication:** findings must be presented in a form that non-technical users can explore.

### 8.3 Role of a Data-Driven Application

A web-based application allows users to load, view, and explore data without writing code. By deploying the application online, results become accessible to mentors, peers, and other stakeholders through a single link. This project addresses the problem by building such an application using Python and Streamlit, supported by a version-controlled GitHub repository.

## 9. Project Objectives

| No. | Objective | Status |
|---|---|---|
| 1 | Understand electric vehicle market data | Partially evidenced: dataset overview confirmed; detailed understanding pending [VERIFY] |
| 2 | Organise and analyse the datasets | Datasets load successfully (confirmed); analysis steps pending verification |
| 3 | Develop meaningful visual analysis | Pending verification of charts in the application and Tableau work |
| 4 | Present findings through a user-friendly interface | Streamlit application exists (confirmed); feature details pending inspection |
| 5 | Deploy the project online | Completed: live Streamlit application and GitHub repository (confirmed) |

## 10. Scope of the Project

| Area | Scope Description |
|---|---|
| Data analytics | Applying descriptive and exploratory techniques to EV data using Python libraries. |
| EV market research | Studying market-related attributes available in the dataset; no external market claims are made. |
| Dataset exploration | Inspecting structure, size, and quality of the training and testing datasets. |

| Area | Scope Description |
|---|---|
| Dashboard and visualisation | Interactive visualisation through the application and, as proposed by the module, Tableau dashboards [PENDING VERIFICATION]. |
| Web-based presentation | Streamlit application deployed online and linked to a GitHub repository. |
| Future enhancements | Additional datasets, richer filtering, advanced statistics, and Tableau integration (see Chapter 23). |

**Out of scope (unless later evidenced):** predictive modelling, real-time data feeds, and external market forecasting. No model, algorithm, or accuracy metric is claimed in this report.

## 11. Tools and Technologies

The table below distinguishes technologies whose use is directly evidenced from those that are part of the module but not yet verified in the implementation.

| Technology | Purpose | Evidence Status |
|---|---|---|
| Python | General-purpose programming language used to write the application logic. | Confirmed (app.py is a Python file; Streamlit apps are written in Python) |
| Streamlit | Framework for building and deploying interactive data web applications. | Confirmed (live Streamlit application) |
| GitHub | Version control and hosting of source code. | Confirmed (public repository, branch main) |
| Web deployment (Streamlit-hosted URL) | Makes the application publicly accessible. | Confirmed (live URL supplied) |
| Pandas | Library for loading, cleaning, and analysing tabular data. | [VERIFY IN app.py] – commonly used for dataset loading, not yet inspected |
| NumPy | Library for numerical computation. | [VERIFY IN app.py] |

| Technology | Purpose | Evidence Status |
|---|---|---|
| Tableau | Visual analytics platform for worksheets, dashboards, and stories. | Listed in the module; **not verified** in implementation or integration |

### 11.1 Purpose of Each Technology

**Python** is widely used in data analytics because of its readable syntax and extensive library ecosystem. **Streamlit** converts Python scripts into web applications with minimal front-end code, which makes it suitable for sharing analytical work. **Pandas** provides the DataFrame structure used to read files, inspect shapes and data types, and filter or aggregate data. **NumPy** supplies fast array operations that underpin many numerical tasks. **Tableau** enables drag-and-drop exploration and the creation of dashboards and narrative stories. **GitHub** stores the source code with history, supports collaboration, and can serve as the source from which the Streamlit application is deployed.

## 12. Data Collection and Extraction

### 12.1 Methodology

Data collection is the process of identifying, obtaining, and storing the data required for analysis. A sound methodology involves: (1) defining the questions the analysis should answer; (2) identifying a suitable and trustworthy source; (3) documenting the source, licence, and date of access; (4) downloading or extracting the data in a structured format such as CSV; and (5) storing the raw files unchanged so that the analysis can be reproduced.

### 12.2 Dataset Source

| Item | Detail |
|---|---|
| Dataset name | [INSERT ACTUAL DATASET NAME] |
| Dataset source / provider | [INSERT ACTUAL DATASET SOURCE] |
| Download link | [INSERT ACTUAL DATASET LINK] |
| Licence / terms of use | [VERIFY LICENCE] |
| Date accessed | [INSERT DATE] |
| File format(s) | [VERIFY FILE FORMAT] |

The dataset source has not been supplied for this report, and no source has been assumed.

### 12.3 Verified Dataset Overview

The deployed application confirms the successful loading of the datasets and displays the following overview:

| Measure | Value (as displayed by the application) |
|---|---|
| Loading message | "Datasets loaded successfully!" |
| Training records | 200,000 |
| Testing records | 100,000 |
| Total columns | 37 |

Column names and data types have not been supplied and are therefore not listed: [ADD COLUMN LIST AFTER INSPECTING THE DATASET].

### 12.4 Importance of Reliable Data Sources

The quality of any analysis is limited by the quality of its data. Reliable sources provide documented collection methods, clear definitions of each column, and known update schedules. Using unverified data can lead to misleading conclusions, and undocumented sources make a project impossible to reproduce or defend during evaluation. For this reason, recording the exact dataset source is listed as a priority item in the pending information checklist in the Appendix.

*(Continued in the next chapters: Data Preparation, Exploratory Analysis, Tableau Workflow, Streamlit Application, Implementation Workflow, and the remaining chapters.)*

## 13. Data Preparation and Cleaning

> **Status: recommended procedures.** No evidence has been supplied that the steps below were executed on the project dataset. They describe good practice and must not be read as completed work until verified.

## 13.1 Recommended Procedures

| Step | Description | Typical Pandas Approach | Status |
|---|---|---|---|
| Check missing values | Count empty or null entries in every column and decide whether to remove, fill, or flag them. | `isnull().sum()` | [VERIFY CLEANING PROCEDURE] |
| Remove duplicates | Identify repeated records that could bias counts and averages. | `duplicated()`, `drop_duplicates()` | [VERIFY CLEANING PROCEDURE] |
| Handle inconsistent values | Standardise spelling, capitalisation, units, and category labels. | `str.strip()`, `str.lower()`, `replace()` | [VERIFY CLEANING PROCEDURE] |
| Check data types | Confirm that numeric, categorical, and date columns are stored correctly. | `dtypes`, `astype()` | [VERIFY CLEANING PROCEDURE] |
| Format data | Apply consistent date formats, numeric precision, and column naming. | `to_datetime()`, `rename()` | [VERIFY CLEANING PROCEDURE] |
| Prepare for analysis | Select relevant columns, create derived fields if needed, and save a clean copy. | `loc[]`, `to_csv()` | [VERIFY CLEANING PROCEDURE] |

## 13.2 Special Considerations for Training and Testing Files

The application loads a training set and a testing set. Whatever cleaning rules are applied should be applied consistently to both files, and the same columns should exist in both. Whether the testing file contains the same 37 columns, or whether any target column is absent from it, has not been verified: [VERIFY COLUMN CONSISTENCY].

### 13.3 Verification Checklist

| Question | Answer |
|---|---|
| Were missing values counted? | [YES / NO / INSERT RESULT] |
| Were duplicates checked? | [YES / NO / INSERT RESULT] |
| Were data types reviewed? | [YES / NO / INSERT RESULT] |
| Were inconsistent categories corrected? | [YES / NO / INSERT RESULT] |
| Is cleaning code present in the repository? | [VERIFY IN GITHUB] |
| Was a cleaned dataset saved? | [YES / NO / INSERT FILE NAME] |

No missing-value counts, duplicate counts, or cleaning results are reported here because none have been supplied.

## 14. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the investigation of a dataset to understand its structure, quality, and patterns before drawing conclusions. This chapter explains how EDA can be performed on an EV dataset. **No actual EDA results are reported.**

### 14.1 Dataset Structure

The first step is to inspect the number of rows and columns, the column names, and the data type of each column. The verified dimensions are 200,000 training records, 100,000 testing records, and 37 columns. Functions such as `shape`, `info()`, and `head()` support this step. [ADD COLUMN SUMMARY AFTER INSPECTION]

### 14.2 Descriptive Statistics

For numeric columns, measures such as count, mean, median, minimum, maximum, and standard deviation summarise the central tendency and spread. For categorical columns, frequency counts show the most and least common categories. In Pandas, `describe()` and `value_counts()` are typical tools.

### 14.3 Distribution Analysis

Histograms and box plots reveal the shape of a variable's distribution, including skewness and outliers. Understanding distributions helps decide whether outliers represent errors or genuine extreme values.

### 14.4 Trend Analysis

If the dataset contains a time-related field such as year, trends can be explored by aggregating a measure over time and plotting it as a line chart. Whether the dataset contains a suitable time field is [VERIFY].

### 14.5 Correlation Analysis

Where several numeric variables exist, a correlation matrix shows the strength and direction of linear relationships between them. Correlation does not imply causation, and results should be interpreted with domain knowledge.

### 14.6 EDA Status

| EDA Activity | Status |
|---|---|
| Structure inspection | Partially evidenced (record and column counts displayed) |
| Descriptive statistics | [PENDING VERIFICATION] |
| Distribution analysis | [PENDING VERIFICATION] |
| Trend analysis | [PENDING VERIFICATION] |
| Correlation analysis | [PENDING VERIFICATION] |

## 15. Data Visualization Using Tableau

> **Status: planned workflow, pending verification.** The module is titled Data Analytics with Tableau. No Tableau workbook, dashboard, story, or screenshot has been supplied, and Tableau has not been confirmed as integrated with the Streamlit application.

### 15.1 Planned Tableau Workflow

| Stage | Description |
|---|---|
| 1. Connect datasets | Open Tableau, choose a connector such as Text file, and load the training and/or testing files. Review the data interpreter and data types on the Data Source page. |

| Stage | Description |
|---|---|
| 2. Select dimensions and measures | Classify fields as dimensions (categorical or descriptive) or measures (numeric, aggregated). Correct any fields Tableau has assigned wrongly. |
| 3. Create worksheets | Build one worksheet per question, placing fields on Rows, Columns, Colour, and Filters shelves. |
| 4. Create charts | Choose appropriate chart types, such as bar charts for comparison, line charts for trends, and scatter plots for relationships. |
| 5. Create dashboards | Combine related worksheets on a dashboard, add filters and actions, and arrange the layout for readability. |
| 6. Create a story | Sequence dashboards or worksheets into a Tableau story, with a caption on each point that explains its meaning. |

## 15.2 Implementation Evidence

| Deliverable | Status | Evidence |
|---|---|---|
| Tableau data connection | [PENDING VERIFICATION] | [ADD SCREENSHOT] |
| Worksheets | [PENDING VERIFICATION] | [ADD TABLEAU WORKSHEET SCREENSHOTS] |
| Dashboard | [PENDING VERIFICATION] | [ADD TABLEAU DASHBOARD SCREENSHOT] |
| Tableau story | [PENDING VERIFICATION] | [ADD TABLEAU STORY SCREENSHOT] |
| Tableau Public / workbook link | [PENDING VERIFICATION] | [INSERT LINK IF PUBLISHED] |

## 15.3 Screenshot Placeholders

> **[ADD TABLEAU DATA SOURCE SCREENSHOT]**
> Figure 15.1: Tableau data source page (to be added after verification).

> **[ADD TABLEAU DASHBOARD SCREENSHOT]**
> Figure 15.2: Tableau dashboard (to be added after verification).

> **[ADD TABLEAU STORY SCREENSHOT]**
> Figure 15.3: Tableau story (to be added after verification).

# 16. Streamlit Web Application Development

This chapter describes only what has been confirmed from the supplied project information.

## 16.1 Confirmed Application Details

| Item | Detail |
|---|---|
| Application titles displayed | Electric Vehicle Market Analysis; EV Market Trends and Insights |
| Main file | `app.py` |
| Branch | `main` |
| Dataset loading message | "Datasets loaded successfully!" |
| Training records | 200,000 |
| Testing records | 100,000 |
| Total columns | 37 |
| Repository | https://github.com/manshikourav04/EV-Market-Analysis |
| Live application | https://manshikourav04-ev-market-analysis-app-twyxno.streamlit.app |

## 16.2 Purpose

The application provides an online interface through which the EV datasets are loaded and summarised, and it is intended to present market trends and insights. Its status messages give immediate feedback that the data files were read correctly.

### 16.3 User Interface and Deployment

The interface opens with the application title and the dataset loading confirmation, followed by a dataset overview showing the record and column counts. Additional interface elements, such as sidebar navigation, filters, tables, or charts, have not been confirmed: [INSPECT THE LIVE APPLICATION AND LIST ADDITIONAL FEATURES]. The application is deployed on Streamlit and publicly reachable through the link above, with its source code in the GitHub repository.

> **[ADD STREAMLIT HOME PAGE SCREENSHOT]**
> Figure 16.1: Application home page showing the title and dataset overview.

## 17. Project Implementation Workflow

The end-to-end workflow below follows the standard analytics lifecycle. Each stage is rated using confirmed evidence only.

```
flowchart LR
  A[Problem identification] --> B[Data collection]
  B --> C[Data preparation]
  C --> D[Analysis]
  D --> E[Visualization]
  E --> F[Application development]
  F --> G[Deployment]
  G --> H[Testing]
  H --> I[Documentation]
```

| Stage | Status | Basis for Status |
|---|---|---|
| 1. Problem identification | Completed | Project title, objectives, and domain are defined. |
| 2. Data collection | Partially completed | Datasets are loaded (confirmed); source and extraction details are unavailable. |
| 3. Data preparation | Pending verification | No cleaning evidence supplied. |
| 4. Analysis | Pending verification | No analytical results supplied. |
| 5. Visualization | Pending verification | Tableau work and in-app charts unverified. |

| Stage | Status | Basis for Status |
|---|---|---|
| 6. Application development | Completed (core) | Streamlit app with `app.py` confirmed; full feature list pending. |
| 7. Deployment | Completed | Live URL and GitHub repository confirmed. |
| 8. Testing | Pending verification | No test evidence supplied. |
| 9. Documentation | In progress | This report; placeholders remain. |

# 18. Dashboard and Application Features

## 18.1 Verified Features

| Feature | Description | Verified |
|---|---|---|
| Application title | Displays "Electric Vehicle Market Analysis" and "EV Market Trends and Insights". | Yes |
| Dataset loading status | Displays "Datasets loaded successfully!". | Yes |
| Dataset overview | Displays training records (200,000), testing records (100,000), and total columns (37). | Yes |
| Public deployment | Accessible through a live Streamlit URL. | Yes |

## 18.2 Features Requiring Inspection

| Possible Feature | Status |
|---|---|
| Data tables or previews | [VERIFY IN LIVE APP] |
| Charts and visualisations | [VERIFY IN LIVE APP] |
| Filters or selection widgets | [VERIFY IN LIVE APP] |
| Sidebar or multi-page navigation | [VERIFY IN LIVE APP] |
| Model or prediction components | [VERIFY IN app.py] – none claimed |

> **[ADD APPLICATION FEATURE SCREENSHOTS]**
> Figure 18.1: Additional application views, once inspected.

# 19. Results and Discussion

## 19.1 Verified Technical Results

| Result | Value |
|---|---|
| Dataset loading | Successful ("Datasets loaded successfully!") |
| Training records | 200,000 |
| Testing records | 100,000 |
| Total records across both sets | 300,000 (sum of the two verified counts) |
| Total columns | 37 |
| Deployment | Live on Streamlit, source on GitHub |

## 19.2 Discussion

The verified results show that the application can load a substantial dataset of 300,000 records across two files and report its basic dimensions. A 2:1 ratio between training and testing record counts is observed, which is a common arrangement when data are split for later modelling, although no modelling is claimed in this project. The project demonstrates a working path from data to a publicly accessible application.

## 19.3 What Is Not Reported

This report makes no claims about EV sales trends, market share, price movements, range, battery characteristics, or other analytical findings, because no such results have been supplied or verified. These should be added only after the analysis is executed and checked: [INSERT VERIFIED FINDINGS].

# 20. Performance Testing

> **Status: all tests pending.** The table lists tests that can be performed. No test evidence has been supplied.

| Test | Method | Expected Observation | Status |
|---|---|---|---|
| Application loading | Open the live URL and note load time and any wake-up delay. | Page renders with the title visible. | [PENDING] |
| Dataset loading | Confirm the loading message and record counts appear. | Message and counts match Chapter 16. | Observed (see Ch. 19); formal test [PENDING] |
| Navigation | Use every page, tab, or link in the app. | No broken views. | [PENDING] |
| Data display | Check tables and charts render fully. | Correct values and labels. | [PENDING] |
| Error handling | Test missing files or unexpected inputs. | Clear messages rather than crashes. | [PENDING] |
| Browser compatibility | Open in Chrome, Edge, Firefox, and Safari. | Consistent layout. | [PENDING] |

A test log with date, tester, browser, device, and result should be added: [INSERT TEST EVIDENCE].

# 21. Web Integration and Deployment

## 21.1 GitHub Repository Management

The source code is stored in the public repository EV-Market-Analysis on the `main` branch, with `app.py` as the main application file. Repository contents beyond `app.py`, such as a README, a requirements file, or dataset files, have not been verified: [VERIFY REPOSITORY CONTENTS].

## 21.2 Streamlit Deployment

The application is deployed at https://manshikourav04-ev-market-analysis-app-twyxno.streamlit.app. The address indicates a Streamlit-hosted deployment connected to the GitHub account. Specific deployment settings, such as Python version, secrets, or resource configuration, have not been supplied and are not stated.

### 21.3 Integration Status

| Integration | Status |
|---|---|
| GitHub to Streamlit deployment | Confirmed by the existence of both links |
| Tableau embedded in the application | **Not confirmed** |
| External data APIs | Not confirmed |

## 22. Challenges and Limitations

### 22.1 General Project Challenges

- Locating a reliable, well-documented EV dataset.
- Handling a dataset of 300,000 records and 37 columns efficiently, including load time on free hosting.
- Ensuring data quality before analysis.
- Choosing visualisations that communicate clearly.
- Combining Tableau and Python-based work into one coherent deliverable.

### 22.2 Issues Actually Encountered

No specific issues encountered during development have been supplied. [INSERT ACTUAL CHALLENGES FACED, IF ANY]

### 22.3 Current Limitations of the Documentation

- Dataset source, column names, and data types are not documented.
- Cleaning, EDA, and testing evidence is unavailable.
- Tableau deliverables are unverified.
- No analytical findings can be reported yet.

## 23. Future Enhancements

| Enhancement | Benefit |
|---|---|
| Improved interactive dashboards | Allow users to explore data without code. |
| Additional EV market datasets | Broaden coverage and enable comparison. |

| Enhancement | Benefit |
| --- | --- |
| More visual analysis | Reveal distributions, trends, and relationships. |
| Tableau integration where appropriate | Embed published Tableau dashboards alongside Python analysis. |
| Improved filtering | Let users focus on specific segments. |
| Advanced statistical analysis | Add rigorous testing of relationships, if justified by the data. |
| Better documentation and testing | Improve reliability and reproducibility. |

## 24. Conclusion

The project set out to analyse electric vehicle market data and present it through an accessible application. The confirmed implementation is a Streamlit application, written in Python in `app.py`, hosted on GitHub, and deployed online. It loads two datasets successfully, reporting 200,000 training records, 100,000 testing records, and 37 columns.

Several parts of the full workflow remain to be evidenced: the dataset source, cleaning steps, exploratory results, Tableau dashboards and story, in-app visualisations, and formal testing. These are documented in this report as recommended methods or pending items rather than as completed work. Once the placeholders are filled with verified information, the report will give a complete and honest account of the project.

## 25. References

1. Python Software Foundation. Python documentation. https://docs.python.org/3/
2. Pandas documentation. https://pandas.pydata.org/docs/
3. NumPy documentation. https://numpy.org/doc/
4. Streamlit documentation. https://docs.streamlit.io/
5. Tableau Help. https://help.tableau.com/
6. GitHub Docs. https://docs.github.com/

Dataset citation: [INSERT ACTUAL DATASET SOURCE]

# 26. Appendix

## A. Project Links and Files

| Item | Value |
|---|---|
| GitHub repository | https://github.com/manshikourav04/EV-Market-Analysis |
| Live application | https://manshikourav04-ev-market-analysis-app-twyxno.streamlit.app |
| Main file | `app.py` |
| Branch | `main` |

## B. Dataset Overview

| Measure | Value |
|---|---|
| Training records | 200,000 |
| Testing records | 100,000 |
| Total columns | 37 |

## C. Screenshot Checklist

- [ ] Streamlit home page with title and loading message
- [ ] Dataset overview section
- [ ] Any charts, tables, or filters in the app
- [ ] GitHub repository main page
- [ ] Tableau data source, worksheets, dashboard, and story
- [ ] Test evidence (browser views)

## D. Pending Information Checklist

- [ ] Institute name, mentor name, and dates
- [ ] Dataset name, source, link, and licence
- [ ] Column names and data types
- [ ] Cleaning steps and results
- [ ] EDA outputs and verified findings

- [ ] Tableau workbook link and screenshots
- [ ] Contents of `app.py` and repository files
- [ ] Testing evidence
- [ ] Issues actually encountered

## E. Suggested Viva Questions and Answers

**Q1. What is the aim of your project?**
To analyse electric vehicle market data and present it through a web application deployed online.

**Q2. Which tools did you use?**
Python and Streamlit for the application, with GitHub for version control and deployment. State Pandas, NumPy, and Tableau use only after verifying them.

**Q3. What does your dataset contain?**
A training set of 200,000 records and a testing set of 100,000 records with 37 columns. Describe the columns after inspecting them.

**Q4. How did you handle data quality?**
Explain only the steps you actually performed, such as checking missing values and duplicates.

**Q5. Why is Streamlit useful?**
It turns Python scripts into shareable web applications with little front-end code.

**Q6. What are the limitations of your work?**
State honestly which parts, such as Tableau dashboards or formal testing, are still pending.

**Q7. What would you improve next?**
Add interactive filters, more datasets, deeper visual analysis, and Tableau integration.